



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 04

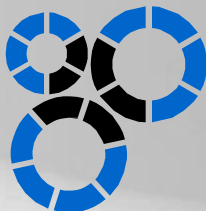
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Outline

- Polyalphabetic Ciphers
 - Playfair Cipher
 - Vigenère Cipher
 - Kasiski Method



◉ Playfair Cipher

- not even the large number of keys in a monoalphabetic cipher provides security
- one approach to improving security was to encrypt multiple letters
- the **Playfair Cipher** is an example
- invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

◉ Playfair Key Matrix

- a 5X5 matrix of letters based on a keyword
- fill in letters of keyword
- fill rest of matrix with other letters
- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

○ Encrypting and Decrypting

- plaintext is encrypted two letters at a time
 1. if a pair is a repeated letter, insert filler like 'X'
 2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end)
 3. if both letters fall in the same column, replace each with the letter below it (wrapping to top from bottom)
 4. otherwise each letter is replaced by the letter in the same row and in the column of the other letter of the pair

◉ Playfair Key Matrix

Plaint Text: SECRET MESSAGE
SE CR ET ME SS AG E
SE CR ET ME S~~X~~ SA GE

Cipher Text: LI DM KL CL XA XB JF

******Decryption process is the same. Just reverse the 4 rules.

◉ Security of Playfair Cipher

- security much improved over monoalphabetic
- since have $26 \times 26 = 676$ digrams
- would need a 676 entry frequency table to analyse (verses 26 for a monoalphabetic)
- and correspondingly more ciphertext
- was widely used for many years
 - eg. by US & British military in WW1
- it can be broken, given a few hundred letters
- since still has much of plaintext structure

◉ Polyalphabetic Ciphers

- Polyalphabetic substitution ciphers
 - improve security using multiple cipher alphabets
 - make cryptanalysis harder with more alphabets to guess and flatter frequency distribution
 - use a key to select which alphabet is used for each letter of the message
 - use each alphabet in turn
 - repeat from start after end of key is reached

◉ Vigenère Cipher

- simplest polyalphabetic substitution cipher
- effectively multiple caesar ciphers
- key is multiple letters long $K = k_1k_2\dots k_d$
- i^{th} letter specifies i^{th} alphabet to use
- use each alphabet in turn
- repeat from start after d letters in message
- decryption simply works in reverse

◉ Example of Vigenère Cipher

- write the plaintext out
- write the keyword repeated above it
- use each key letter as a caesar cipher key
- encrypt the corresponding plaintext letter
- eg. using keyword *deceptive*
- key: deceptivedeceptivedeceptive
- plaintext: wearediscoveredsaveyourself
- ciphertext: **ZICVTWQNGRZGVTWAVZHCQYGLMGJ**

Example of Vigenère Cipher

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

More Example of Vigenère Cipher

- Key: LEMON
- Plain Text: SECRET MESSAGE (13)
- Key-stream: LEMONL EMONLEM (13)
-
- Encryption: Cross-match S,L-> D; E,E->I; C,M-> O....
- Cipher Text: DIOFRE QQGFLKQ
- Decryption: Start with key-stream and look for the corresponding cipher text in the same row/column.

◉ Aid for encryption/decryption

- simple aid can assist with en/decryption
- a **Saint-Cyr Slides** a simple manual aid
- a slide with repeated alphabet
- line up plaintext 'A' with key letter, eg 'C'
- then read off any mapping for key letter
- can bend round into a **cipher disk**
- or expand into a **Vigenère Table**

◉ Security of Vigenère Ciphers

- have multiple ciphertext letters for each plaintext letter
- hence letter frequencies are obscured
- but not totally lost
- start with letter frequencies
- see if look monoalphabetic or not
- if not, then need to determine number of alphabets, since then can attach each

◉ Kasiski Method

- method developed by Babbage / Kasiski
- repetitions in ciphertext give clues to period
- so find same plaintext an exact period apart
- which results in the same ciphertext
- of course, could also be random fluke
- eg. repeated “VTW” in previous example
- suggests size of 3 or 9
- then attack each monoalphabetic cipher individually using same techniques as before



Thank You

